Question 50

05: Improve Developer Productivity

How can developers safely generate realistic synthetic data sets (e.g., JSON mock objects or SQL seed scripts) for local development using GitHub Copilot?

A. By uploading production database dumps into the prompt window.

✔ **B. By providing a target schema or interface sample in a comment and prompting Copilot to generate mock records following that structure.**

C. By connecting Copilot directly to production SQL databases.

D. Copilot cannot generate mock structured data formats like JSON or SQL.

Correct Answer: B

Copilot excels at generating synthetic mock data. Providing a schema interface definition or single example record in a comment allows Copilot to produce structured JSON arrays or SQL `INSERT` statements populated with mock data.

Option Breakdown

- **A. Uploading production dumps:** Incorrect. Exposing production PII or live credentials violates security best practices.
- **B. Schema definition + mock generation prompt:** Correct. Generates clean synthetic test datasets matching structural types without exposing real user data.
- **C. Direct production database connection:** Incorrect. Copilot has no direct database driver or connection layer.
- **D. Cannot generate structured formats:** Incorrect. Copilot easily outputs JSON, YAML, SQL, and CSV mock data formats.

💡 **GH-300 Exam Tip**

Generating synthetic data via Copilot protects user privacy by eliminating the bad habit of downloading real PII or customer database dumps to local developer environments.